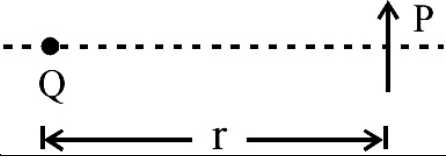
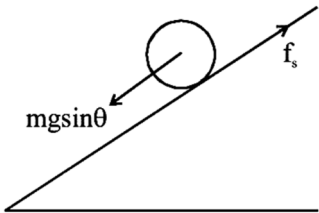
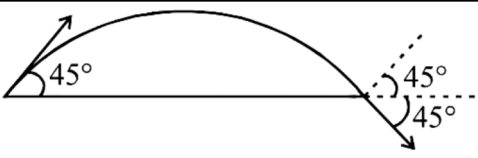
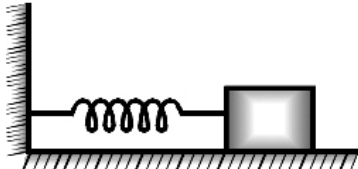
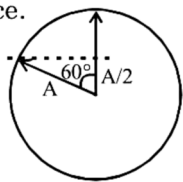
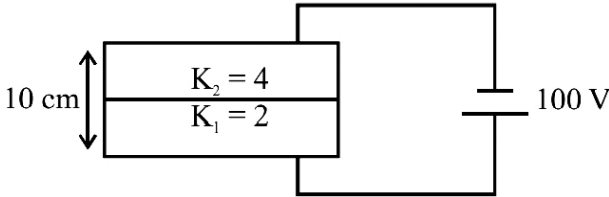
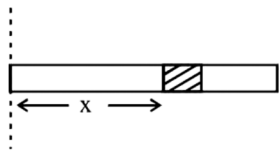


1.	A point charge Q and an electric dipole of dipole moment P are placed as given in figure. The force on dipole due to point charge is $\frac{kPQ}{r^N}$. The value of N is:- 	C
	1	
	2	
	3	
	4	
Sol.	Force on dipole due to point charge is same as force on point charge due to dipole so $F = \frac{kP}{r^3} \times Q = \frac{kPQ}{r^3}$ so $N = 3$	
2.	A uniform disk of mass m and radius R rolls, without slipping, down a fixed plane inclined at an angle 30° to the horizontal. The linear acceleration of the disk (in m/sec^2) is	B
	5 m/s^2	
	$\frac{10}{3} \text{ m/s}^2$	
	$\frac{20}{3} \text{ m/s}^2$	
	None of these	
Sol.	$mg \sin \theta - f_s = ma$ $f_s R = \frac{mR^2}{2} \alpha$ $a = R\alpha$ $a = \frac{2g \sin \theta}{3}$ $= \frac{2 \times 10 \times \frac{1}{2}}{3} = \frac{10}{3} \text{ m/s}^2$ 	
3.	A particle is projected from ground at an angle such that it covers maximum horizontal distance on ground. In this case the angle between initial and final velocity vector is	C
	$\frac{\pi}{6}$	
	$\frac{\pi}{4}$	
	$\frac{\pi}{2}$	
	Depend on speed of projection	
Sol.		
4.	During free expansion of an ideal gas under adiabatic condition, the internal	D

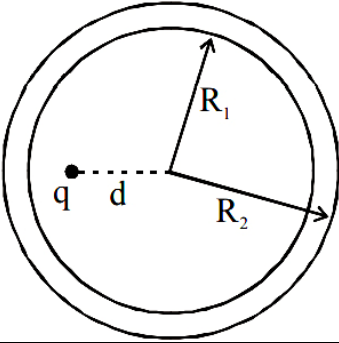
	energy of the gas	
	Decreases	
	Initially decreases and then increases	
	Increases	
	Remains constant	
Sol.	$\Delta Q = 0$ for free expansion $\Delta W = 0$ so $\Delta U = 0$	
5.	2 kg of ice at -20°C is mixed with 5 kg of water at 20°C . The water content of final mixture is (Latent heat of ice = 80 kcal/kg, specific heat of water = 1 kcal $\text{kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$, specific heat of ice = 0.5 kcal $\text{kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$)	B
	7 kg	
	6 kg	
	4 kg	
	3 kg	
Sol.	$\Delta Q_{\text{ice}} (-20^{\circ} \rightarrow 0^{\circ}\text{C}) = 2 \times \frac{1}{2} \times 20 = 20 \text{ Kcal}$ $\Delta Q_{\text{water}} (20^{\circ} \rightarrow 0^{\circ}\text{C}) = 5 \times 1 \times 20 = 100 \text{ Kcal}$ $\Delta Q_{\text{water}} > \Delta Q_{\text{ice}}$ so ice melts and we can see 80 Kcal of heat is remaining so 1 kg ice melts so water content 6kg.	
6.	In a reference frame K two particles move along the X-axis, one of mass m_1 with velocity $\vec{v}_1$ and the other of mass m_2 with velocity $\vec{v}_2$. There exists another reference frame K_0 in which the total kinetic energy of the two masses m_1 and m_2 will be the minimum. The instantaneous velocity $\vec{v}$ of the reference frame K_0 in the reference frame K will be	D
	$\vec{v} = \vec{v}_1 + \vec{v}_2$	
	$\vec{v} = \vec{v}_1$	
	$\vec{v} = \vec{v}_2$	
	$\vec{v} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$	
Sol.	K_0 must be COM frame.	
7.	A compound microscope consists of an objective lens of focal length 1cm and an eye piece of focal length 5 cm with a separation of 10 cm. The distance between an object and the objective lens, at which the strain on the eye is minimum is	B
	$\frac{3}{4}$	
	$\frac{5}{4}$	
	$\frac{2}{3}$	
	None of these	

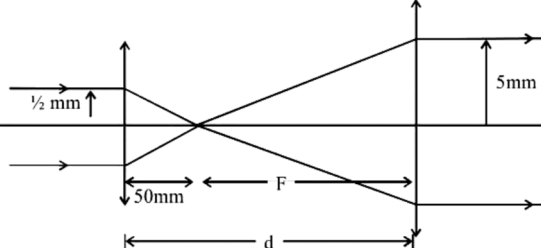
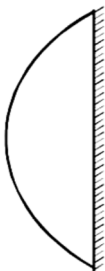
Sol.	For minimum strain on eye final image must be at infinite. $v_f \rightarrow \infty$ so $u_e = f_e = 5\text{cm}$ so v_0 (image distance of objective) = 5 cm $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for objective $\frac{1}{+5} - \frac{1}{u_0} = \frac{1}{1}$ $\frac{1}{u_0} = \frac{1}{5} - 1 = -\frac{4}{5}$ $u_0 = -\frac{5}{4}$ $u_0 = -\frac{50}{40}\text{cm}$ $n = 50\text{ cm}$	
8.	Spring of spring constant 1200 Nm^{-1} is mounted on a smooth frictionless surface and attached to a block of mass 3 kg. Block is pulled 2 cm to the right and released. The time taken by block cover a distance of 1 cm after release is 	D
	5 sec	
	0.83 sec	
	0.01 sec	
	0.052 sec	
Sol.	$\omega = 20\text{ rad/s}$ $T = \frac{2\pi}{\omega} = \frac{\pi}{10}$ time taken to travel 1 cm distance. $\omega t = \frac{\pi}{3}$ $t = \frac{\pi}{3} = \frac{\pi}{3 \times 20} = \frac{\pi}{60} = 0.052\text{s}$ 	
9.	Two planets P_1 and P_2 having masses M_1 and M_2 revolve around the Sun in elliptical orbits, with time periods T_1 and T_2, respectively. The minimum and maximum distances of planet P_1 from the Sun are R and $3R$, respectively, whereas for planet P_2 these are $2R$ and $4R$, respectively, where R is a constant. Assuming M_1 and M_2 are much smaller than the mass of the Sun, the magnitude of $\frac{T_2}{T_1}$ is	C
	$\frac{2}{3} \sqrt{\frac{2M_1}{3M_1}}$	

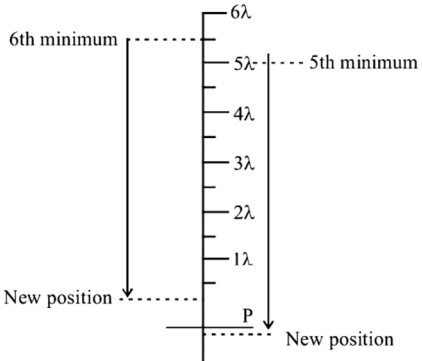
	$\frac{3}{2} \sqrt{\frac{3M_2}{2M_1}}$	
	$\frac{3}{2} \sqrt{\frac{3}{2}}$	
	$\frac{2}{3} \sqrt{\frac{2}{3}}$	
Sol.	$T^2 = \frac{4\pi^2}{GM_s} a^3$ $a = \frac{R_{\max} + R_{\min}}{2}$ $\frac{T_2^2}{T_1^2} = \left(\frac{3R}{2R}\right)^3 \quad \frac{T_2}{T_1} = \left(\frac{3}{2}\right)^{3/2}$	
10.	In a parallel plate capacitor the distance between the plates is 10 cm. Two dielectric slabs of thickness 5 cm each and dielectric constant $K_1 = 2$ and $K_2 = 4$ respectively, are inserted between the plates. A potential of 100 V is applied across the capacitor as shown in the figure. The value of the net bound surface charge density at the interface of the two dielectric is 	A
	$-\frac{2000}{3} \epsilon_0$	
	$-\frac{1000}{3} \epsilon_0$	
	$-250\epsilon_0$	
	$\frac{2000}{3} \epsilon_0$	
Sol.	$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{2C \times 4C}{6C} = \frac{4}{3} C$ $Q = \frac{4C}{3} V = \frac{4 \epsilon_0 A}{3 \times 5 \times 10^{-2}} \times 100$ $\sigma = \frac{Q}{A} = \frac{4 \times 100 \times 100}{3 \times 5} \epsilon_0 = \frac{8000 \epsilon_0}{3}$ $\sigma_b = +\sigma \left(1 - \frac{1}{k_1}\right) - \sigma \left(1 - \frac{1}{k_2}\right)$ $= -\sigma \left(\frac{1}{k_1} - \frac{1}{k_2}\right) = -\sigma \left(\frac{4-2}{4 \times 2}\right) = \frac{-2\sigma}{8} = \frac{-\sigma}{4}$ $= -\frac{2000 \epsilon_0}{3}$	
11.	From the point of view of significant figures which of the following statement is	C

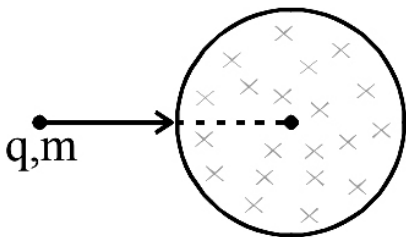
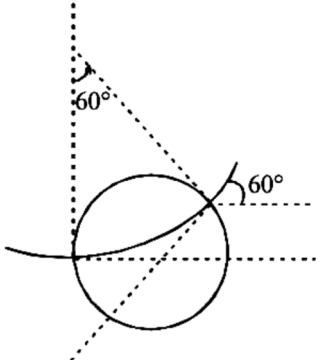
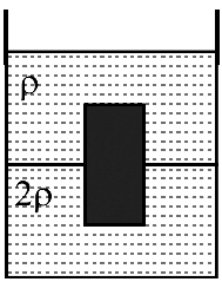
	correct? (i) $10.2 \text{ cm} + 8 \text{ cm} = 18.2 \text{ cm}$ (ii) $2.53 \text{ m} - 1.2 \text{ m} = 1.33 \text{ m}$ (iii) $4.2 \text{ m} \times 1.4 \text{ m} = 5.88 \text{ m}^2$ (iv) $3.6 \text{ m} / 1.75 \text{ sec} = 2.1 \text{ m/sec}$	
	(i) & (iv) only	
	(ii) & (iii) only	
	(iv) only	
	(ii) & (iv) only	
Ans	(iv) only	
12.	The mass per unit length of a rod (length 2 m) varies as $\rho = 3x \text{ kg/m}$. The moment of inertia (in kg m^2) of the rod about a perpendicular axis passing through the tip of the rod (at $x = 0$) is	B
	10	
	12	
	14	
	16	
Sol.	$dI = [(\rho) dx] x^2$  $\int dI = 3 \int_0^2 x^3 dx$ $I = 3 \left(\frac{2^4}{4} \right) = 12 \text{ kgm}^2$	
13.	A steel rod of length 5 m is fixed between two rigid supports. The coefficient of linear expansion of steel is $12.5 \times 10^{-6}/^\circ\text{C}$. The stress in the rod for an increase in temperature of 40°C is $x \times 10^8 \text{ N/m}^2$. (Young's modulus for steel is $2 \times 10^{11} \text{ N/m}^2$). The value of x is	B
	2.5	
	1	
	1.5	
	$\frac{2}{3}$	
Sol.	$\frac{F}{A} = Y \alpha \Delta T$ $= 2 \times 10^{11} \times 12.5 \times 10^{-6} \times 40$ $= 25 \times 40 \times 10^5 = 1 \times 10^8 \text{ N/m}^2$	
14.	In a meter bridge circuit as shown, when one more resistance of 100Ω is connected in parallel with unknown resistance x, then ratio $\frac{\ell_1}{\ell_2}$ become 2. If ℓ_1 is balance length of the uniform wire AB, then the value of x must be:	B

	50Ω	
	100Ω	
	200Ω	
	400Ω	
Sol.	$\frac{l_1}{l_2} = \frac{100}{\left(\frac{x \times 100}{100 + x}\right)} = 2$ $100 + x = 2x$ $x = 100\Omega$	
15.	A small loop of wire of area $A = 0.01 \text{ m}^2$, $N = 40$ turns and resistance $R = 20\Omega$ is initially kept in a uniform magnetic field B in such a way that the field is normal to the plane of the loop. When it is pulled out of the magnetic field, a total charge of $Q = 2 \times 10^{-5} \text{ C}$ flows through the coil. The magnitude of the field B is	A
	$1 \times 10^{-3} \text{ T}$	
	$4 \times 10^{-3} \text{ T}$	
	Zero	
	Unobtainable, as the data is insufficient	
Sol.	$Q = \frac{\Delta\phi}{R} = \frac{NBA}{R}$ $\text{so } B = \frac{QR}{NA} = \frac{2 \times 10^{-5} \times 20}{40 \times 0.01} = 1 \times 10^{-3} \text{ T}$	
16.	An LR circuit with a battery is connected at $t = 0$. Which of the following quantity is not zero just after the connection?	D
	Current in the circuit	
	Magnetic field energy in the inductor	
	Power delivered by the battery	
	Emf induced in the inductor	
Sol.	At $t = 0$ LR Circuit can be seen as So $i = 0$ $P = P_{\text{Bat}} = 0$ $U_{\text{inductor}} = 0$ $V_L = E$	
17.	A hollow, conducting spherical shell of inner radius R_1 and outer radius R_2	D

	encloses a charge q inside, which is located at a distance d ($< R_1$) from the centre of the spheres. The potential at the centre of the shell is	
		
	Zero	
	$\frac{1}{4\pi\epsilon_0 d}$	
	$\frac{1}{4\pi\epsilon_0} \left(\frac{q}{d} - \frac{q}{R_1} \right)$	
	$\frac{1}{4\pi\epsilon_0} \left(\frac{q}{d} - \frac{q}{R_1} + \frac{q}{R_2} \right)$	
Sol.	$V = V_{\text{due to } q} + V_{\text{due to induce charge at inner surface}} + V_{\text{due to charge on outer surface}}$ $= \frac{kq}{d} + \left(-\frac{kq}{R_1} \right) + \left(\frac{kq}{R_2} \right)$	
18.	A p-type semiconductor has acceptor levels 57 meV above the valence band. The maximum wavelength of light required to create a hole is (Planck's constant $h = 6.6 \times 10^{-34} \text{ J-s}$)	C
	57 Å	
	$57 \times 10^{-3} \text{ Å}$	
	217100 Å	
	$11.61 \times 10^{-3} \text{ Å}$	
Sol.	$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{57 \times 10^{-3} \times 1.6 \times 10^{-19}} = 217100 \text{ Å}$	
19.	A collimated beam of light of diameter 1 mm is propagating along the x-axis. The beam is to be expanded to a collimated beam of diameter 10 mm using a combination of two convex lenses. A lens of focal length of 50 mm and another lens with focal length F are to be kept at a distance d between them. The value of F and d respectively, are	D
	450 mm and 10 mm	
	400 mm and 500 mm	
	550 mm and 600 mm	
	500 mm and 550 mm	

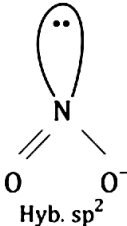
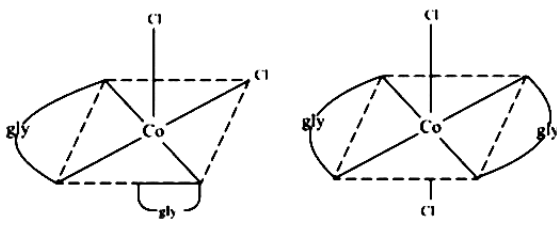
Sol.	 $\frac{1/2}{50} = \frac{5}{F}$ $F = 500 \text{ mm}$ $d = 550 \text{ mm}$	
20.	A 20 cm long string, having a mass of 1.0 g, is fixed at both the ends. The tension in the string is 0.5 N. The string is set into vibrations using an external vibration of frequency 100 Hz. The separation (in cm) between the successive nodes on the string is	C
	3 cm	
	4 cm	
	5 cm	
	None of these	
Sol.	$T = 0.5 \text{ N}$ $\mu = \frac{m}{l} = 5 \times 10^{-3} \text{ kg/m}$ $v = \sqrt{\frac{T}{\mu}} = 10 \text{ m/s}$ $\lambda = \frac{v}{f} = 0.1 \text{ m} = 10 \text{ cm}$ distance between two successive nodes $\frac{\lambda}{2} = 5 \text{ cm}$	
21.	A plano convex lens of refractive index 1.5 and radius of curvature 30 cm is silvered on its flat surface. The distance (in cm) an object be placed relative to the silvered lens so as to form a real image of the same size as the object is	60
Sol.	Focal length of silvered lens. $\frac{1}{f_{\text{eq}}} = \frac{-2(\mu - 1)}{R} \quad n-1\left(\frac{1}{R}\right)$ $f_{\text{eq}} = -\frac{f}{2} = -30 \text{ cm}$ $-\frac{1}{f_{\text{eq}}} = 2\left(\frac{n-1}{R}\right)$ $f_{\text{eq}} = -\frac{R}{2(n-1)}$ and to get image of same size object must be on centre of curvature. 	

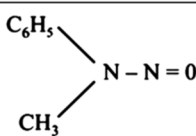
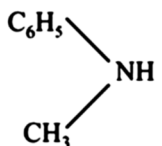
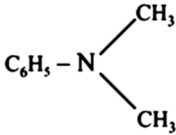
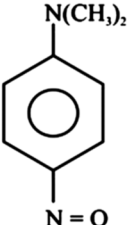
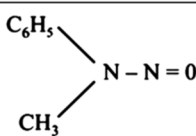
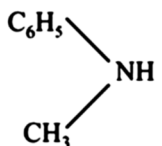
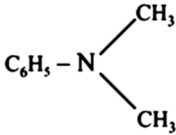
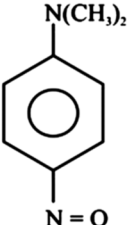
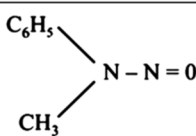
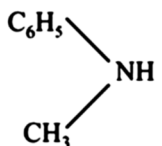
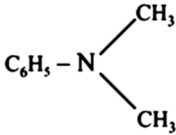
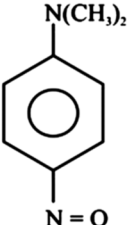
22.	In a Young's experiment the upper slit is covered by a thin glass plate of refractive index 1.4 while the lower slit is covered by another glass plate, having the same thickness as the first one but having refractive index 1.7. Interference pattern is observed using light of wavelength $5400 \text{ \AA}$. It is found that the point P on the screen where the central maximum ($n = 0$) fell before the glass plates were inserted now has $3/4$ the original intensity. It is further observed that what used to be the fifth maximum earlier, now lies below the point P while the sixth minimum lies above P. The thickness of glass plate (in μm) is Answer to the nearest integer. (Absorption of light by glass plate may be neglected):-	9
Sol.	The optical path difference developed due to insertion of two glass plates is $x = (\mu_2 - \mu_1)t = (1.7 - 1.4)t = 0.3t$ where t is the thickness of each glass plate. Phase difference $\delta = \frac{2\pi}{\lambda}x$. The intensity distribution equation is $I = I_0 \cos^2\left(\frac{\delta}{2}\right)$ At P $\frac{3}{4}I_0 = I_0 \cos^2\frac{\delta}{2}$  New position or $\cos\frac{\delta}{2} = \pm\frac{\sqrt{3}}{2}$ or $\frac{\delta}{2} = n\pi \pm \frac{\pi}{6}$ or $\delta = 2n\pi \pm \frac{\pi}{3}$ The corresponding path difference is $n\lambda \pm \frac{1}{6}\lambda.$ After insertion of glass plates the 5th maximum goes below the point P and the 6th minimum lies above the point P. Thus due to insertion of glass plates the change of path difference is $(5 + \epsilon)\lambda$ where $\epsilon = 0$. Again the 6th minimum lies above P. The change in path difference is $\left(5 + \frac{1}{2} - \eta\right)\lambda, \eta > 0$ Thus, $(5 + \epsilon)\lambda = \left(5 + \frac{1}{2} - \eta\right)\lambda$ $\epsilon + \eta = \frac{1}{2}$ or $\epsilon < \frac{1}{2}$ The change of optical path must be $\left(5 + \frac{1}{6}\right)\lambda.$ Thus, $0.3t = \left(5 + \frac{1}{6}\right)\lambda = \frac{31\lambda}{6}$ or $t = \frac{31 \times 5400 \times 10^{-10}}{6 \times 0.3} \text{ m} = 9.3 \times 10^{-6} \text{ m}$	
23.	A particle having mass m, charge q enters a cylinder region having uniform magnetic field B in the inward direction as shown. If the particle is deviated by 60° as it emerges out of the field then the time spent by it in the field is $\frac{\pi m}{NqB}$. The value of N is:-	3

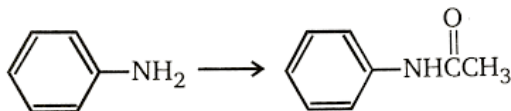
		
Sol.	$\frac{\pi}{3} = \omega t$ $\frac{\pi}{3} = \omega t$ $t = \frac{\pi m}{3qB}$ 	
24.	A cylinder of uniform cross-section is immersed into two liquids as shown in figure. Equal volume of cylinder is submerged in both the liquids and density of cylinder is ρ. The magnitude of acceleration of the cylinder at the shown instant is g/K. The value of K is:- 	2
Sol.	$a = \frac{\rho \frac{V}{2} g + 2\rho \frac{V}{2} g - \rho V g}{\rho V} = \frac{g}{2}$ $K = 2$	
25.	A long wire PQR is made by joining two wires PQ and QR of equal radii. PQ has length 4.8 m and mass 0.06 kg. QR has length 2.56 m and mass 0.2 kg. The wire PQR is under a tension of 80N. A sinusoidal wave pulse of amplitude 3.5 cm is sent along the wire PQ from the end P. No power is dissipated during the propagation of the wave pulse. The time taken by the wave pulse to reach the other end R is x (sec). The value of $100x$ is	14
Sol.	$V_{PQ} = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{80}{0.06/4.8}} = 80 \text{ m/s}$ $V_{QR} = \sqrt{\frac{80}{0.2/2.56}} = 32 \text{ m/s}$ $t = \frac{4.8}{80} + \frac{2.56}{32} = 0.14 \text{ s}$	
26.	A solution of (–)-1-chloro-1-phenylethane in toluene racemises slowly in the presence of a small amount of SbCl_5 due to the formation of Free radical	C

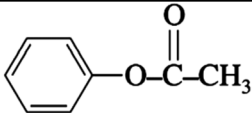
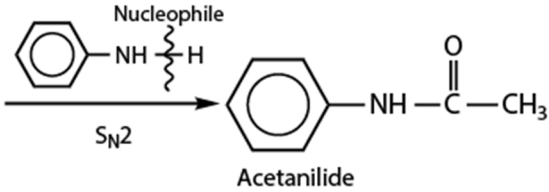
	Carbene	
	Carbocation	
	Carbanion	
Sol.	$\text{Cl}-\underset{\text{Ph}}{\text{CH}}-\text{CH}_3 \xrightarrow[\text{Toluene}]{\text{SbCl}_5} \text{Ph}-\overset{\oplus}{\text{CH}}-\text{CH}_3 + \text{SbCl}_6^-$ (carbocation) $\downarrow$ $\text{Ph}-\underset{\text{Cl}}{\text{CH}}-\text{CH}_3 + \text{SbCl}_5$ (d + l) mixture i.e., Racemic mixture	
27.	An organic compound A upon reacting with NH_3 given B. On heating, B given C. C in presence of KOH reacts with Br_2 to give $\text{CH}_3\text{CH}_2\text{NH}_2$. A is	D
	$\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$	
	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	
	CH_3COOH	
	$\text{CH}_3\text{CH}_2\text{COOH}$	
Sol.	$\underset{\text{A}}{\text{CH}_3\text{CH}_2\text{COOH}} + \text{NH}_3 \rightarrow \underset{\text{B}}{\text{CH}_3\text{CH}_2\text{CO}_2\text{NH}_4}$ $\xrightarrow{\Delta} \underset{\text{C}}{\text{CH}_3\text{CH}_2\text{CONH}_2}$ $\xrightarrow{\text{KOH} + \text{Br}_2 \text{ (Hoffmann Bromamide reaction)}} \text{CH}_3\text{CH}_2\text{NH}_2 + \text{H}_2\text{O}$	
28.	In a Cu-voltameter, mass deposited in 30s is m gm. If the time-current graph is shown in the following figure What is the electrochemical equivalent of Cu?	B
	$m/2$	
	$m/3$	
	$m/4$	
	$\frac{m}{63.5}$	

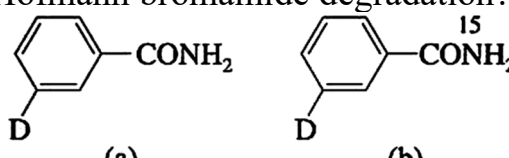
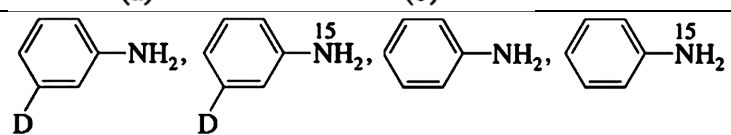
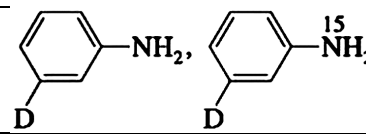
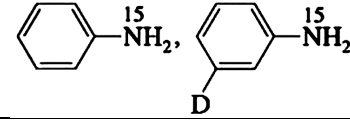
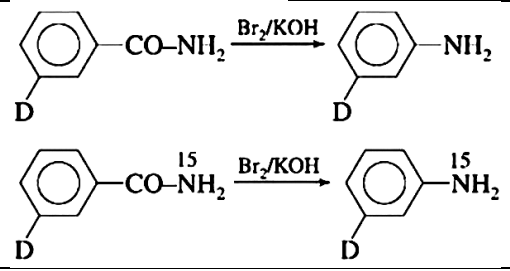
Sol.	Total charge passed in 30 sec. $= \frac{1}{2} \times 100 \times 10 + 100 \times 10 + 100 \times 10 + \frac{1}{2} \times 100 \times 10$ $= 3 \times 100 \times 10 \text{mA} \times \text{s}$ $= 3000 \times 10^{-3} \text{amp s} = 3\text{C}$ $W = ZQ$ So $Z = \frac{W}{Q} = \frac{m}{3}$	
29.	Choose the correct statement with respect to the bond length of CO and CO ⁺	A
	The bond length in CO is 1.128 Å and CO ⁺ is 1.115 Å because during conversion of CO to CO ⁺ , electron is removed from anti bonding orbital.	
	The bond length in CO is 1.115 Å and CO ⁺ is 1.128 Å because during conversion of CO to CO ⁺ electron is removed from bonding orbital.	
	During conversion of CO to CO ⁺ bond length does not vary because bond order remain same.	
	The bond length in CO is 1.115 Å and CO is 1.128 Å because bond order decreases during conversion of CO to CO ⁺	
Sol.	Because of the greater difference in electronegativity between C and O, the energy of the molecular orbital in CO are in the following order, $\sigma_{1s}^2 \sigma_{2s}^{*2} \sigma_{2s}^2 \left[\pi_{2py}^2 = \pi_{2pz}^2 \right] \sigma_{2z}^2 \sigma_{2s}^{*2}$ Bond order in CO = $\frac{1}{2} (10 - 4) = 3$ When CO is converted to CO⁺, one electron is removed from antibonding molecular orbital. Hence, the bond order in CO⁺ is greater than CO.	
30.	20% of N ₂ O ₄ molecules are dissociated in a sample of gas at 27°C and 760 torr. Mixture has the density at equilibrium equal to	D
	1.48 g/L	
	1.84 g/L	
	2.25 g/L	
	3.12 g/L	
Sol.	The reaction is $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ $n = 2$ and $\alpha = \frac{20}{100} = 0.2$ $D = 4.6$, initial vapour density d = vapour density at equilibrium $\alpha = \frac{D-d}{(n-1)d}$ $0.2 = \frac{46-d}{(2-1)d}$ $d = 38.3$ Molar mass equilibrium $= 2 \times 38.3 = 76.6$ $PM = dRT$ Here, d = density of gas mixture $d(\text{mix}) = \frac{PM}{RT} = \frac{1 \times 76.6}{0.0821 \times 300}$ $= 3.12 \text{ g/L}$	

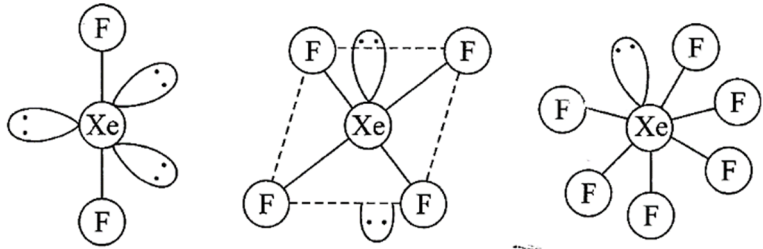
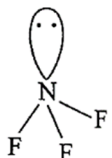
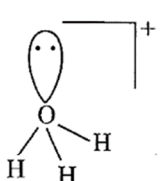
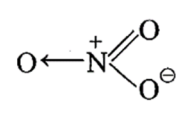
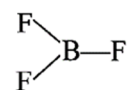
31.	Two oxides of nitrogen, NO and NO ₂ are allowed to react together at 243 K and form coloured compound of nitrogen (X). When compound (X) reacts with water to yield another compound of nitrogen (Y). The shape of the anion (Y) molecule is	A
	Angular	
	Triangular pyramidal	
	Tetrahedron	
	Square planar	
Sol.	Here X is N₂O₃ (light blue coloured) and Y is Nitrous (HNO₂) acid $\text{NO (g)} + \text{NO}_2\text{(g)} \xrightarrow{243 \text{ K}} \underset{\substack{(\text{x}) \\ \text{(Blue coloured solid)}}}{\text{N}_2\text{O}_3}$ $\xrightarrow{\text{H}_2\text{O}} \underset{(\text{y})}{\text{HNO}_2}$ Anion of HNO₂ is NO₂⁻ and it is a angular or bent in shape  Hyb. sp²	
32.	The octahedral complex/complex ion which shows both facial and meridional isomers is	C
	Triglycinato cobalt (III)	
	Tris (ethylene diamine) cobalt (III)	
	Dichloro diglycinato cobalt (III)	
	Trioxalate cobaltate (III)	
Sol.	Complexes of the type [M(aa)₂a₂] show geometrical or fac-mer isomerism. (where, aa = bidentate ligand and a is monodentate ligand.) Among the given, dichlorodiglycinato cobalt (III), i.e., [CoCl₂(gly)₂] is such salt, so it will show fac and mer isomerism. 	
33.	The correct code for stability, of oxidation states for given cations is I. Pb⁺² > Pb⁺⁴, Tl⁺ < Tl⁺³ II. Bi⁺³ < Sb⁺³, Sn⁺² < Sn⁺⁴ III. Pb⁺² > Pb⁺⁴, Bi⁺³ > Bi⁺⁵ IV. Tl⁺³ < In⁺³, Sn⁺² > Sn⁺⁴ V. Sn⁺² < Pb⁺², Sn⁺⁴ > Pb⁺⁴ VI. Sn⁺² < Pb⁺², Sn⁺⁴ < Pb⁺⁴	C
	V and VI	

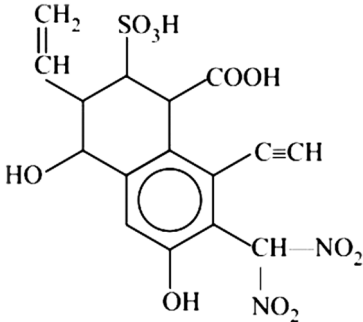
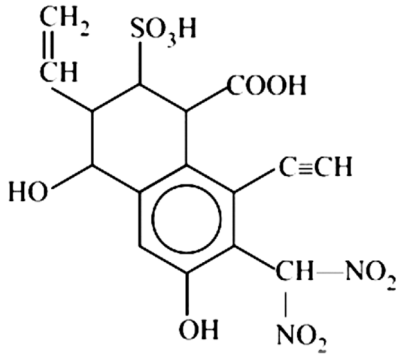
	I, II and VI																					
	III and V																					
	II and IV																					
Sol.	Due to inert effect the stability of lower oxidation state gradually increases while stability of higher oxidation state gradually decreases down the group in elements of group 13th to 15th. So correct order are III. $\text{Pb}^{2+} > \text{Pb}^{4+}$, $\text{Bi}^{3+} > \text{Bi}^{5+}$ IV. $\text{Sn}^{2+} < \text{Pb}^{2+}$, $\text{Sn}^{4+} < \text{Pb}^{4+}$																					
34.	Match list – with list – II and select the correct answer using the codes given below <table><thead><tr><th></th><th>List - I</th><th></th><th>List-II</th></tr></thead><tbody><tr><td>(P)</td><td>$\text{C}_6\text{H}_5 - \text{NH}_2 \xrightarrow{\text{NaNO}_2/\text{HCl}}$</td><td>1.</td><td></td></tr><tr><td>(Q)</td><td> $\xrightarrow{\text{NaNO}_2/\text{HCl}}$</td><td>2.</td><td>$(\text{CH}_3 - \text{CH}_2 - \text{CH}_2) - \overset{\oplus}{\text{N}} - \text{N} = \text{O}$</td></tr><tr><td>(R)</td><td>$(\text{CH}_3 - \text{CH}_2 - \text{CH}_2)_3 - \text{N} \xrightarrow{\text{NaNO}_2/\text{HCl}}$</td><td>3.</td><td>$\text{C}_6\text{H}_5 - \overset{\oplus}{\text{N}} = \text{NCl}^\ominus$</td></tr><tr><td>(S)</td><td> $\xrightarrow{\text{NaNO}_2/\text{HCl}}$</td><td>4.</td><td></td></tr></tbody></table>		List - I		List-II	(P)	$\text{C}_6\text{H}_5 - \text{NH}_2 \xrightarrow{\text{NaNO}_2/\text{HCl}}$	1.		(Q)	 $\xrightarrow{\text{NaNO}_2/\text{HCl}}$	2.	$(\text{CH}_3 - \text{CH}_2 - \text{CH}_2) - \overset{\oplus}{\text{N}} - \text{N} = \text{O}$	(R)	$(\text{CH}_3 - \text{CH}_2 - \text{CH}_2)_3 - \text{N} \xrightarrow{\text{NaNO}_2/\text{HCl}}$	3.	$\text{C}_6\text{H}_5 - \overset{\oplus}{\text{N}} = \text{NCl}^\ominus$	(S)	 $\xrightarrow{\text{NaNO}_2/\text{HCl}}$	4.		A
	List - I		List-II																			
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(S)	 $\xrightarrow{\text{NaNO}_2/\text{HCl}}$	4.																				
	P(3), Q (1), R(2), S(4)																					
	P(3), Q(1), R(4), S(2)																					
	P(1), Q(3), R(4), S(2)																					
	P(4), Q(3), R(2), S(1)																					

Sol.	Here the correct matching is P(3), Q(1), R(2), S(4) $\text{C}_6\text{H}_5-\text{NH}_2 \xrightarrow{\text{NaNO}_2/\text{HCl}} \text{C}_6\text{H}_5-\overset{\bullet}{\text{N}}=\text{NCl}^\oplus$ $\begin{array}{c} \text{C}_6\text{H}_5 \\ \diagup \\ \text{NH} \\ \diagdown \\ \text{CH}_3 \end{array} \xrightarrow{\text{NaNO}_2/\text{HCl}} \begin{array}{c} \text{C}_6\text{H}_5 \\ \diagup \\ \text{N}-\text{N}=\text{O} \\ \diagdown \\ \text{CH}_3 \end{array}$ $(\text{CH}_3-\text{CH}_2-\text{CH}_3)_2-\text{N} \xrightarrow{\text{NaNO}_2/\text{HCl}} (\text{CH}_3-\text{CH}_2-\text{CH}_3)-\overset{\bullet}{\text{N}}-\text{N}=\text{O}$ $\begin{array}{c} \text{CH}_3 \\ \diagup \\ \text{C}_6\text{H}_5-\text{N} \\ \diagdown \\ \text{CH}_3 \end{array} \xrightarrow{\text{NaNO}_2/\text{HCl}} \begin{array}{c} \text{N}(\text{CH}_3)_2 \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{N}=\text{O} \end{array}$																									
35.	Determine the correct order of hybridization of the central atom in the following species NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3	B																								
	dsp^2 , dsp^3 , sp^2 and sp^3																									
	sp^3 , dsp^2 , dsp^3 , sp^2																									
	dsp^2 , sp^2 , sp^3 , dsp^3																									
	dsp^2 , sp^3 , sp^2 , dsp^3																									
Sol.	Hybridization of N in NH_3 is sp^3 that of Pt in $[\text{PtCl}_4]^{2-}$ is dsp^2 that P in PCl_5 is sp^3d and that of B in BCl_3 is sp^2 .																									
36.	Initially, 0.8 mol of PCl_5 and 0.2 mol of PCl_3 are mixed in a one litre vessel. At equilibrium, 0.4 mol of PCl_3 is present. The value of K_c for the reaction, $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$, would be	A																								
	0.13 mol L^{-1}																									
	0.66 mol L^{-1}																									
	0.013 mol L^{-1}																									
	0.05 mol L^{-1}																									
Sol.	<table><tr><td></td><td>$\text{PCl}_5(\text{g})$</td><td>$\rightleftharpoons$</td><td>$\text{PCl}_3(\text{g})$</td><td>$+$</td><td>$\text{Cl}_2(\text{g})$</td></tr><tr><td>Initial concentration (mol L^{-1})</td><td>0.8</td><td></td><td>0.2</td><td></td><td>0</td></tr><tr><td>Equilibrium concentration</td><td>$(0.8 - x)$</td><td></td><td>0.4</td><td></td><td>x</td></tr><tr><td>Inc. in concentration of PCl_3</td><td colspan="5">$= 0.4 - 0.2 = 0.2$</td></tr></table> So $x = 0.2$ $K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} = \frac{0.4 \times 0.2}{0.6}$ $= 0.13 \text{ mol L}^{-1}$		$\text{PCl}_5(\text{g})$	$\rightleftharpoons$	$\text{PCl}_3(\text{g})$	$+$	$\text{Cl}_2(\text{g})$	Initial concentration (mol L^{-1})	0.8		0.2		0	Equilibrium concentration	$(0.8 - x)$		0.4		x	Inc. in concentration of PCl_3	$= 0.4 - 0.2 = 0.2$					
	$\text{PCl}_5(\text{g})$	$\rightleftharpoons$	$\text{PCl}_3(\text{g})$	$+$	$\text{Cl}_2(\text{g})$																					
Initial concentration (mol L^{-1})	0.8		0.2		0																					
Equilibrium concentration	$(0.8 - x)$		0.4		x																					
Inc. in concentration of PCl_3	$= 0.4 - 0.2 = 0.2$																									
37.	All but one of the following compounds reacts with aniline to give acetanilide. Which one does not ?	C																								
	 Aniline Acetanilide																									

	$\text{CH}_3\text{C}(=\text{O})\text{Cl}$	
	$\text{H}_3\text{C}-\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{CH}_3$	
	$\text{CH}_3\text{C}(=\text{O})\text{H}$	
		
Sol.	$\text{CH}_3-\text{C}(=\text{O})-\text{Cl}$ Leaving group $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{CH}_3$ Leaving group $\text{CH}_3-\text{C}(=\text{O})-\text{O}-\text{C}_6\text{H}_5$ Leaving group Nucleophile  $\text{S}_{\text{N}}2$ Acetanilide Hence $\text{CH}_3 - \text{CH} = \text{O}$ does not give acetanilide with aniline	
38.	An Alkyl bromide reacts with sodium to form 4, 5-diethyloctane. The alkyl bromide is	D
	$\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{Br}$	
	$\text{CH}_3(\text{CH}_2)_4\text{CH}_2\text{Br}$	
	$\text{CH}_3(\text{CH}_2)_3\text{CHBrCH}_3$	
	$\text{CH}_3(\text{CH}_2)_2\text{CHBrCH}_2\text{CH}_3$	
Sol.	$\text{R} - \text{X} \xrightarrow{\text{Na/ether}} \text{R} - \text{R}$ So given alkane divided by half and gets corresponding alkyl halide as following :	
39.	The decreasing order of – I effect of the following is : I. $-\text{NH}_3^+$ II. $-\text{NO}_2$ III. $-\text{CN}$ IV. $-\text{COOH}$	A
	I > II > III > IV	
	II > I > III > IV	
	I > II > III > IV	
	II > I > IV > III	
40.	Which test is used to differentiate between aldose from ketose	B
	Molisch test	
	Seliwanoff test	
	Barfoed's test	
	Ninhydrin test	

41.	What is the true regarding complete combustion of gaseous isobutene	D
	$\Delta H = \Delta E$	
	$\Delta H > \Delta E$	
	$\Delta H = \Delta E = 0$	
	$\Delta H < \Delta E$	
Sol.	$C_4H_{10}(g) + 6.5 O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$ $\Delta n = [4 - 7.5] = -3.5$ $\Delta H = \Delta E + \Delta n_g RT$ $\therefore \Delta H < \Delta E$	
42.	Which is not a pair of reducing sugars-	A
	Glucose, sucrose	
	Fructose, lactose	
	Glucose, maltose	
	Glucose, fructose	
43.	What are the constituent amines formed when the mixture (a) and (a) undergoes Hofmann bromamide degradation?	B
	 (a) (b)	
		
		
		
		
Sol.		
44.	In XeF_2 , XeF_4 and XeF_6 the number of lone pair of electrons on Xe respectively is	C
	1, 2, 3	
	2, 3, 1	
	3, 2, 1	
	4, 1, 1	

Sol.	The structures of XeF_2, XeF_4 and XeF_6 is given as  In $\text{XeF}_2 \rightarrow 2bp + 3lp$, In $\text{XeF}_4 \rightarrow 4bp + 2lp$ In $\text{XeF}_6 \rightarrow 6bp + 1lp$	
45.	Among the following, the correct isostructural pairs are NF_3, NO_3^-, BF_3, H_3O^+, HN_3 [NF_3, NO_3^-] and [BF_3, H_3O^+] [NF_3, HN_3] and [NO_3^-, BF_3] [NF_3, H_3O^+] and [NO_3^-, BF_3] [NF_3, H_3O^+] and [HN_3, BF_3]	C
Sol.	Structure of NF_3  $H = \frac{5 + 3}{2} = 4$ Hybridisation = sp^3, according to VSEPR, shape of NF_3 is trigonal pyramidal. Structure of H_3O^+  $H = \frac{6 + 3 - 1}{2} = 4$ Hybridisation = sp^3, shape = trigonal pyramidal Structure of NO_3^-  $H = \frac{5 + 1}{2} = 3$ Hybridisation = sp^2, shape = trigonal planar Structure of BF_3  $H = \frac{3 + 3}{2} = 3$ Hybridisation = sp^2 shape = trigonal planar.	

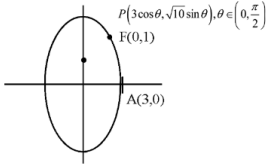
	Structure of N_3H $H-N \equiv N \equiv N$ The hybridisation of central N-atom is sp, but due to irregular distribution of electron cloud its structure is slightly bent. Hence, option (c) is correct regarding isostructural pairs. Note In case of NO_3^-, π-bond is not considered in structural determination of molecule/ion.	
46.	How many acidic H – atoms are present in this compound that can react with $R^\ominus$ for $R-MgX$ to give alkane 	6
Sol.	Here 6 H-atoms are acidic enough to react with $R^\ominus$ for $R-MgX$. Here hydrogen atom of $-SO_3H$, $-COOH$, $-C \equiv CH$, $-OH$ (phenolic and alcoholic both) and $-CH-NO_2$ NO_2 are acidic. 	
47.	A current of 2 amp when passed for 5 hour through a molten salt deposits 22.2 g of metal of atomic mass 177. The oxidation state of the metal salt is [Report your answer by rounding it upto nearest whole number]	3
Sol.	$W = \frac{ItE}{96500}$ $E = \frac{W \times 96500}{It} = \frac{22.2 \times 96500}{2 \times 5 \times 3600} = 59.5$ $\text{Equivalent mass} = \frac{\text{Atomic mass}}{\text{Oxidation state}}$ $59.5 = \frac{177}{n}$ So $n \approx 3$.	
48.	How many isomers of $C_5H_{11}Br$ show stereochemical inversion when it undergoes SN^2 reaction?	4

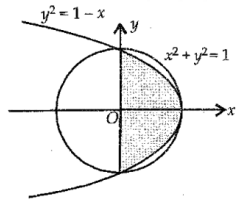
Sol.	Total 8 structural isomers are possible for $C_5H_{11}Br$, out of which two isomer have Br at chiral carbon which can show stereochemical inversion. $CH_3CH_2CH_2CH_2(Br)CH_3$ and its enantiomer; $(CH_3)_2CHCH(Br)CH_3$ and its enantiomer	
49.	Calculate the pH of solution when 100 mL, 0.1 M CH_3COOH and 100 mL, 0.1 M $HCOOH$ are mixed together. (Given : $K_a(CH_3COOH) = 2 \times 10^{-5}$, $K_a(HCOOH) = 6 \times 10^{-5}$) (Multiply your answer by 10)	27
Sol.	As the volume is doubled, concentration would be half. The total $[H^+]$ is obtained by addition of $[H^+]$ from both acids. $[H^+] = \sqrt{(K_{a1}C_1 + K_{a2}C_2)}$ $= \sqrt{(2 \times 10^{-5} \times \frac{0.1}{2} + 6 \times 10^{-5} \times \frac{0.1}{2})}$ $= 2 \times 10^{-3}$ pH = 2.7	
50.	How many of the following gives flame test : Li, Na, K, Be, Mg, Ca, Ba, Cs	6
51.	The mean and the median of the following ten numbers in increasing order 10, 22, 26, 29, 34, x, 42, 67, 70, y are 42 and 35 respectively, then $\frac{y}{x}$ is equal to	C
	9/4	
	7/2	
	7/3	
	8/3	
Sol.	(c) : Given, mean = 42 $\Rightarrow \frac{10+22+26+29+34+x+42+67+70+y}{10} = 42$ $\Rightarrow 300 + x + y = 420 \Rightarrow x + y = 120 \quad \dots(i)$ Also, median = 35 $\Rightarrow \frac{34+x}{2} = 35 \Rightarrow x = 36$ From (i), we get $y = 120 - 36 = 84$ Now, $\frac{y}{x} = \frac{84}{36} = \frac{7}{3}$	
52.	The value of $\cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$ is	B
	$\frac{3}{4} + \cos 20^\circ$	
	$\frac{3}{4}$	
	$\frac{3}{2}(1 + \cos 20^\circ)$	
	$\frac{3}{2}$	

Sol.	$\cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$ $= \left(\frac{1 + \cos 20^\circ}{2} \right) + \left(\frac{1 + \cos 100^\circ}{2} \right) - \frac{1}{2} (2 \cos 10^\circ \cos 50^\circ)$ $= 1 + \frac{1}{2} (\cos 20^\circ + \cos 100^\circ) - \frac{1}{2} [\cos 60^\circ + \cos 40^\circ]$ $= \left(1 - \frac{1}{4} \right) + \frac{1}{2} [\cos 20^\circ + \cos 100^\circ - \cos 40^\circ]$ $= \frac{3}{4} + \frac{1}{2} [2 \cos 60^\circ \times \cos 40^\circ - \cos 40^\circ] = \frac{3}{4}$	
53.	$\lim_{t \rightarrow 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \dots + n^{\frac{1}{\sin^2 t}} \right)^{\sin^2 t}$ is equal to	B
	$n^2 + n$	
	n	
	$\frac{n(n+1)}{2}$	
	n^2	
Sol.	$\lim_{t \rightarrow 0} \left(1^{\frac{1}{\sin^2 t}} + 2^{\frac{1}{\sin^2 t}} + \dots + n^{\frac{1}{\sin^2 t}} \right)^{\sin^2 t}$ $= \lim_{t \rightarrow 0} \left(1^{\csc^2 t} + 2^{\csc^2 t} + \dots + n^{\csc^2 t} \right)^{\sin^2 t}$ $= \lim_{t \rightarrow 0} n \left(\left(\frac{1}{n} \right)^{\csc^2 t} + \left(\frac{2}{n} \right)^{\csc^2 t} + \dots + 1 \right)^{\sin^2 t} = n(0 + 0 + 0 + \dots + 1) = n$	
54.	Let A, B, C be 3×3 matrices such that A is symmetric and B and C are skew-symmetric. Consider the statements (S1) $A^{13}B^{26} - B^{26}A^{13}$ is symmetric (S2) $A^{26}B^{13} - C^{13}A^{26}$ is symmetric Then	A
	Only S2 is true	
	Only S1 is true	
	Both S1 and S2 are false	
	Both S1 and S2 are true	
Sol.	Given, $A^T = A, B^T = -B, C^T = -C$ Let $M = A^{13}B^{26} - B^{26}A^{13}$ Then, $M^T = (A^{13}B^{26} - B^{26}A^{13})^T$ By using property $(AB)^T = B^T A^T$ $= (A^{13}B^{26})^T - (B^{26}A^{13})^T = B^{26}A^{13} - A^{13}B^{26}$ Hence, M is skew symmetric Let, $N = A^{26}C^{13} - C^{13}A^{26}$ Then, $N^T = (A^{26}C^{13})^T - (C^{13}A^{26})^T = -(C^{13})^T (A^{26})^T + A^{26}C^{13} = N$ Hence, N is symmetric, $\therefore$ Only S2 is true.	

55.	If the system of equations $x + y + z = 6$ $2x + 5y + \alpha z = \beta$ $x + 2y + 3z = 14$ has infinitely many solutions, then $2\alpha + \beta$ is equal to	D
	8	
	36	
	44	
	52	
Sol.	Given system of equations are $x + y + z = 6$ $2x + 5y + \alpha z = \beta$ $x + 2y + 3z = 14$ From the given equations. $x + y = 6 - z \quad \dots(i)$ $x + 2y = 14 - 3z \quad \dots(ii)$ Subtract (i) from (ii), $\Rightarrow y = 8 - 2z, \text{ then } x = z - 2$ Now, put the values x & y In eq. $2x + 5y + \alpha z = \beta$ $2(z - 2) + 5(8 - 2z) + \alpha z = \beta$ $(\alpha - 8)z = \beta - 36$ For having infinite solutions $\alpha - 8 = 0 \text{ \& } \beta - 36 = 0 \Rightarrow \alpha = 8, \beta = 36$ Required sum $= 2\alpha + \beta = 52$	
56.	For $a, b > 0$, let $f(x) = \begin{cases} \frac{\tan((a+1)x) + b \tan x}{x}, & x < 0 \\ 3, & x = 0 \\ \frac{\sqrt{ax + b^2 x^2} - \sqrt{ax}}{b\sqrt{a}(x\sqrt{x})}, & x > 0 \end{cases}$ be a continuous function at $x = 0$. Then $\frac{b}{a}$ is equal to	D
	5	
	4	
	8	
	6	

Sol.	$f(x)$ is continuous at $x = 0$ $\Rightarrow \lim_{x \rightarrow 0} f(x) = f(0) = 3$ $\lim_{x \rightarrow 0^+} \frac{\sqrt{ax + b^2 x^2} - \sqrt{ax}}{b\sqrt{ax}\sqrt{x}} = 3$ By Rationalization we get $\lim_{x \rightarrow 0^+} \frac{ax + b^2 x^2 - ax}{b\sqrt{ax}^{\frac{3}{2}}(\sqrt{ax + b^2 x^2} + \sqrt{ax})}$ $\lim_{x \rightarrow 0^+} \frac{b^2}{b\sqrt{a}(\sqrt{a + b^2 x} + \sqrt{a})}$ $\frac{b}{\sqrt{a} \cdot 2\sqrt{a}} \Rightarrow \frac{b}{2a} = 3 \Rightarrow \frac{b}{a} = 6$	
57.	Let $f(x) = 3 \sin^4 x + 10 \sin^3 x + 6 \sin^2 x - 3, x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$ Then, f is	D
	Increasing $\left(-\frac{\pi}{6}, \frac{\pi}{2}\right)$	
	Decreasing in $\left(0, \frac{\pi}{2}\right)$	
	Increasing in $\left(-\frac{\pi}{6}, 0\right)$	
	Decreasing in $\left(-\frac{\pi}{6}, 0\right)$	
Sol.	Given that $f(x) = 3 \sin^4 x + 10 \sin^3 x + 6 \sin^2 x - 3, x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$ $f'(x) = 12 \sin^3 x \cos x + 30 \sin^2 x \cos x + 12 \sin x \cos x$ $= 6 \sin x \cos x (2 \sin^2 x + 5 \sin x + 2)$ $= 6 \sin x \cos x (2 \sin x + 1)(\sin x + 2) = 0$ $\Rightarrow x = -\frac{\pi}{6}, 0, \frac{\pi}{2}$ Decreasing in $\left(-\frac{\pi}{6}, 0, \frac{\pi}{2}\right)$ $\begin{array}{c} - \qquad \qquad + \\ \text{-----} \text{-----} \\ -\frac{\pi}{6} \qquad 0 \qquad \qquad \frac{\pi}{2} \end{array}$	
58.	Let complex number z_1, z_2 satisfy $Re(z_1) > 0, Re(z_2) > 0$ and $Re(z_1^2) = Re(z_2^2) = 2(Re(z))$ represents the real part of a complex number z . then the minimum value of $ z_1 + 2 + \bar{z}_2 + 2 - \bar{z}_1 - z_2 $ is _____.	C
	$\sqrt{2}$	
	$3\sqrt{2}$	
	$4\sqrt{2}$	
	$5\sqrt{2}$	

Sol.	Let Complex number z_k correspond to point $P_k(x_k, y_k), K = 1, 2$ Let P_1, P_2 be symmetric about the x-axis, then P_1, P_2 are on the right branch of the hyperbola $C: x^2 - y^2 = 2$. Let F_1, F_2 be the foci of $C, F_1(-2, 0), F_2(2, 0)$ By def. of hyp. $ P_1 F_1 = P_1 F_2 + 2\sqrt{2}, P_2 F_1 = P_2 F_2 + 2\sqrt{2}$ $= z_1 + 2 + \overline{z_2} + 2 - \overline{z_1} - z_2 $ $= z_1 + 2 + \overline{z_2} + 2 - z_1 - \overline{z_2} $ $= P_1 F_1 + P_2 F_1 - P_1 P_2 $ $= 4\sqrt{2} + P_1 F_2 + P_2 F_2 - P_1 P_2 $ $= 4\sqrt{2}.$	
59.	In the Cartesian plane, ellipse 'C' is given by $\frac{x^2}{9} + \frac{y^2}{10} = 1$. Point F is the upper focus of C, A is the right vertex C, and P is a moving point 'C' in the first quadrant. The maximum area of quadrilateral OAPF is	B
	$3\sqrt{11}$	
	$\frac{3\sqrt{11}}{2}$	
	$\sqrt{11}$	
	$\frac{\sqrt{11}}{2}$	
Sol.	 Then, $S_{OAPF} = S_{\triangle OAP} + S_{\triangle OFP} = \frac{1}{2} \cdot 3 \cdot \sqrt{10} \sin \theta + \frac{1}{2} \cdot 1 \cdot 3 \cos \theta$ $= \frac{3}{2} (\sqrt{10} \sin \theta + \cos \theta) = \frac{3\sqrt{11}}{2} \sin(\theta + \phi)$ $\phi = \tan^{-1} \left(\frac{1}{\sqrt{10}} \right)$ When $\theta = \arctan \sqrt{10}$ $\therefore$ quadrilateral OAPF has Maximum area as $\frac{3\sqrt{11}}{2}$.	
60.	The area of the region described by $A = \{(x, y): x^2 + y^2 \leq 1 \text{ and } y^2 \leq 1 - x\}$ is	D
	$\frac{\pi}{2} - \frac{4}{3}$	
	$\frac{\pi}{2} - \frac{2}{3}$	
	$\frac{\pi}{2} + \frac{2}{3}$	
	$\frac{\pi}{2} + \frac{4}{3}$	

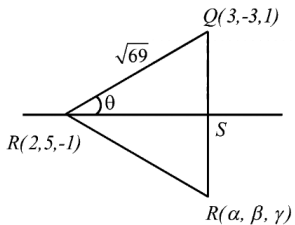
Sol.	$\text{Area} = \frac{\pi}{2} + 2 \int_0^1 \sqrt{1-x} dx$  $= \frac{\pi}{2} - 2 \cdot \frac{(1-x)^{3/2}}{3/2} \Big _0^1 = \frac{\pi}{2} - \frac{4}{3} \left[(1-x)^{3/2} \right]_0^1 = \frac{\pi}{2} + \frac{4}{3}$	
61.	Let $A = \{x_1, x_2, \dots, x_7\}$ and $B = \{y_1, y_2, y_3\}$ be two sets containing seven and three distinct elements respectively. Then the total number of functions $f: A \rightarrow B$ that are onto, if there exist exactly three elements x in A such that $f(x) = y_2$, is equal to	D
	14. 7C_2	
	16. 7C_3	
	12. 7C_2	
	14. 7C_3	
Sol.	Number of ways of selection of three elements in A such that $f(x) = y_2$ is 7C_3 . Now for remaining 4 elements in A , we have 2 elements in B $\therefore$ Total number of onto functions $= {}^7C_3 \times (2^4 - {}^4C_1(2-1)^4) = {}^7C_3 \times 14$	
62.	Let the solution curve $y = y(x)$ of the differential equation $\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp \left\{ \frac{x^3 - \tan^{-1} x^3}{\sqrt{1+x^6}} \right\}$ (here exp represents exponential function) Pass through the origin. Then $y(1)$ is equal to	B
	$\exp \left\{ \frac{4+\pi}{4\sqrt{2}} \right\}$	
	$\exp \left\{ \frac{4-\pi}{4\sqrt{2}} \right\}$	
	$\exp \left\{ \frac{1-\pi}{4\sqrt{2}} \right\}$	
	$\exp \left\{ \frac{\pi-4}{4\sqrt{2}} \right\}$	

Sol.	Given differentiate equation is $\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp\left\{\frac{x^3 - \tan^{-1} x^3}{\sqrt{1+x^6}}\right\},$ Which is in the form $\frac{dy}{dx} + Py = Q$ Here, $P = -\frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}}, Q = 2x \exp\left\{\frac{x^3 - \tan^{-1} x^3}{\sqrt{1+x^6}}\right\}$ Let $I = -\int \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} dx$ Put $x^3 = \tan \theta \Rightarrow 3x^2 dx = \sec^2 \theta d\theta$ $I = -\int \frac{\tan \theta \cdot \theta}{\sec^3 \theta} \times \sec^2 \theta d\theta = -\int \theta \cdot \sin \theta d\theta$ $= -\left[\theta \int \sin \theta d\theta - \int \left(\int \sin \theta d\theta\right) d\theta\right]$ $= -\left[-\theta \cdot \cos \theta + \sin \theta\right] = -\left[\frac{x^3}{\sqrt{1+x^6}} - \tan^{-1}(x^3) \cdot \frac{1}{\sqrt{1+x^6}}\right]$ $\therefore I.F = e^{\int P dx} = e^I$ Solution of the D.E. is $y \cdot e^I = \int e^I \cdot 2x \cdot \exp\left\{\frac{x^3 - \tan^{-1}(x^3)}{\sqrt{1+x^6}}\right\} dx$ $\int 2x \cdot e^{\frac{-x^3 + \tan^{-1}(x^3)}{\sqrt{1+x^6}}} \cdot e^{\frac{x^3 - \tan^{-1}(x^3)}{\sqrt{1+x^6}}} dx$ $= \int 2x \cdot dx = x^2 + x \quad \therefore ye^I = x^2 + c$ $\therefore y = x^2 \exp\left\{\frac{-\tan^{-1}(x^3) + x^3}{\sqrt{1+x^6}}\right\} + c \exp\left\{\frac{x^3 - \tan^{-1}(x^3)}{\sqrt{1+x^6}}\right\}$ This curve passes through the origin. So, $0 = 0 + c \Rightarrow c = 0$ The required solution is $y(x) = x^2 \exp\left\{\frac{x^3 - \tan^{-1}(x^3)}{\sqrt{1+x^6}}\right\}$ At $x = 1$ $y(1) = \exp\left(\frac{1 - \frac{\pi}{4}}{\sqrt{2}}\right) = \exp\left(\frac{4 - \pi}{4\sqrt{2}}\right)$	
63.	Four points having position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar and $(\sin \alpha)\vec{a} + (2\sin 2\beta)\vec{b} + (3\sin 3\gamma)\vec{c} - \vec{d} = \vec{0}$, then the least value of $\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma$ is $\frac{1}{14}$ 14 6 $\frac{1}{\sqrt{6}}$	A

Sol.	Since four points having position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are co-planar So $\sin \alpha + 2 \sin 2\beta + 3 \sin 3\gamma = 1$ Let $\vec{p} = \hat{i} + 2\hat{j} + 3\hat{k}$ $\vec{q} = \sin \alpha \hat{i} + \sin 2\beta \hat{j} + \sin 3\gamma \hat{k}$ $\vec{p} \cdot \vec{q} \leq \vec{p} \vec{q}$ $\Rightarrow \sin \alpha + 2 \sin 2\beta + 3 \sin 3\gamma \leq \sqrt{14} \cdot \sqrt{\sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma}$ $\Rightarrow \sin^2 \alpha + \sin^2 2\beta + \sin^2 3\gamma \geq \frac{1}{14}$	
64.	If $f(x)$ satisfies the relation $f\left(\frac{5x-3y}{2}\right) = \frac{5f(x)-3f(y)}{2}, \forall x, y \in R$ and $f(0) = 3, f'(0) = 2$, then the fundamental period of $\sin(f(x))$ is	A
	π	
	2π	
	3π	
	4π	
Sol.	$f\left(\frac{5x-3y}{2}\right) = \frac{5f(x)-3f(y)}{2}$ $\Rightarrow f\left(\frac{5x-3y}{5-3}\right) = \frac{5f(x)-3f(y)}{2}$ So $f(x)$ must be a linear function $f(x) = ax + b \rightarrow f(x) = 2x + 3$ Period of $\sin(2x + 3)$ is π	
65.	If from each of the three boxes containing 3 white and 1 black, 2 white and 2 black, 1 white and 3 black balls, one ball is drawn at random, then the probability that 2 white and 1 black ball will be drawn is	A
	$\frac{13}{32}$	
	$\frac{1}{4}$	
	$\frac{1}{32}$	
	$\frac{3}{16}$	
Sol.	You can have W, W, B or W, B, W or B, W, W Required probability $= \frac{3}{4} \cdot \frac{2}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{2}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{2}{4} \cdot \frac{1}{4} = \frac{9}{32} + \frac{3}{32} + \frac{1}{32} = \frac{13}{32}$	
66.	Let $F: [3, 5] \rightarrow R$ be a twice differential function on $(3, 5)$ such that $F(x) = e^{-x} \int_3^x (3t^2 + 2t + 4F'(t)) dt$. Then $F(x) =$ _____	C
	$\frac{x^3 - x^2 - 36}{e^x + 4}$	
	$\frac{x^3 + x^2 - 36}{e^x + 4}$	
	$\frac{x^3 + x^2 - 36}{e^x - 4}$	
	$\frac{x^3 + x^2 + 36}{e^x - 4}$	

Sol.	$e^x F(x) = \int_3^x (3t^2 + 2t + 4F^1 t) dt$ Let $y = F(x)$ $\frac{dy}{dx} + \frac{e^x}{e^x - 4} y = \frac{3x^2 + 2x}{e^x - 4}$ I.F = $e^x - 4$ Solution $y(e^x - 4) = \int (3x^2 + 2x) dx$ $y(e^x - 4) = x^3 + x^2 + c$ $F(3) = 0 \Rightarrow y(3) = 0$ $c = -36 \quad F(x) = \frac{x^3 + x^2 - 36}{e^x - 4}$	
67.	The true solution set of $\log_{\tan \frac{\pi}{6}} \left(\frac{x^2 - x + 1}{x + 2} \right) > -2$ is equal to	A
	$-1 < x < 5$	
	$-2 < x < 5$	
	$-\infty < x < -2 \cup -1 < x < 5$	
	$x \geq 5$	
Sol.	$\log_{\frac{1}{\sqrt{3}}} \left(\frac{x^2 - x + 1}{x + 2} \right) > -2$ $\Rightarrow 0 < \frac{x^2 - x + 1}{x + 2} < 3$ Since $\Rightarrow 0 < \frac{x^2 - x + 1}{x + 2} < 3$, hence domain is $x + 2 > 0$ Now $\frac{x^2 - x + 1}{x + 2} < 3$ $x^2 - x + 1 > 3x + 6$ $\Rightarrow x^2 - 4x - 5 < 0$ $\Rightarrow (x - 5)(x + 1) < 0$ Hence $x \in (-1, 5)$ also domain is $x > -2$ $\Rightarrow -1 < x < 5$	
68.	Unit vector $\vec{c}$ is inclined at an angle θ to unit vectors $\vec{a}$ and $\vec{b}$ which are perpendicular. If $\vec{c} = \lambda(\vec{a} + \vec{b}) + \mu(\vec{a} \times \vec{b})$ where λ, μ be real, then θ belongs to	D
	$\left(-\frac{\pi}{4}, 0\right)$	
	$\left[0, \frac{\pi}{4}\right)$	
	$\left[\frac{3\pi}{4}, \pi\right)$	
	$\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$	

Sol.	$\vec{a} \cdot \vec{b} = 0$ $\vec{c} \cdot \vec{a} = \vec{c} \cdot \vec{b} = \cos \theta$ $\vec{c} \cdot \vec{a} = \lambda (\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b}) + \mu \vec{a} \cdot (\vec{a} \times \vec{b})$ $\cos \theta = \lambda$ $c \cdot c = \lambda^2 (a+b)^2 + \mu^2 (1.1 \sin 90^\circ)^2 + 2\lambda\mu \{(\vec{a} + \vec{b}) \cdot (\vec{a} \times \vec{b})\}$ $1 = \lambda^2 (1+1+2.0) + \mu^2 + 2\lambda\mu(0) \quad \left[\because \vec{a} =1, \vec{b} =1 \text{ and } \vec{a} \cdot \vec{b} = 0 \right]$ $1 = 2\cos^2 \theta + \mu^2$ $\mu^2 = 1 - 2\cos^2 \theta \geq 0$ $-\frac{1}{\sqrt{2}} \leq \cos \theta \leq \frac{1}{\sqrt{2}} \Rightarrow \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$	
69.	Let R be the set of the real numbers. Statements-1: $A = \{(x, y) \in R \times R : y - x \text{ is an integer}\}$ is an equivalence relation on R. Statements-2: $B = \{(x, y) \in R \times R : x = \alpha y \text{ for some rational number } \alpha\}$ is an equivalence relation on R.	A
	Statement-1 is true, Statement-2 is false.	
	Statement-1 is false, Statement-2 is true.	
	Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.	
	Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.	
Sol.	(a) : $y - x = \text{integer}$ and $z - y = \text{integer} \Rightarrow z - x = \text{integer}$ $\therefore (x, y) \in A \text{ and } (y, z) \in A$ $\Rightarrow (x, z) \in A \Rightarrow$ Transitive Also $(x, x) \in A$ is true $\Rightarrow$ Reflexive As $(x, y) \in A \Rightarrow (y, x) \in A \Rightarrow$ Symmetric Hence A is an equivalence relation but B is not. Since, $(0, y)$ is in B but $(y, 0)$ is not in B.	
70.	Let A and B be two square matrices of order 3 such that $A = 3$ and $B = 2$. Then $A^T A (\text{adj}(2A))^{-1} (\text{adj}(4B)) (\text{adj}(AB))^{-1} A A^T$ is equal to	C
	108	
	32	
	64	
	81	

Sol.	(c) : We have $ A^T A (\text{adj } (2A))^{-1} (\text{adj}(4B)) (\text{adj}(AB))^{-1} A A^T $ $= A^T A \frac{1}{ \text{adj } 2A } \text{adj } 4B \frac{1}{ \text{adj } AB } A A^T $ $= A ^2 \frac{1}{(2^2)^3 \text{adj } A } (4^2)^3 \text{adj } B \frac{1}{ \text{adj } AB } A ^2$ [$\because A = A^T$ and $\text{adj } kA = k^{n-1} \text{adj } A$ where n is order of A] $= A ^2 \frac{1}{2^6 A ^2} 4^6 B ^2 \frac{1}{ A ^2 B ^2} A ^2$ [$\because \text{adj } A = A ^{n-1}$ where n is order of A] $= \frac{4^6}{2^6} = 64$	
71.	Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(3, -3, 1)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to	81
Sol.	 $RQ = \sqrt{1 + 64 + 4} = \sqrt{69}$ $\overrightarrow{RQ} = \hat{i} - 8\hat{j} + 2\hat{k}$ Vector along $\overrightarrow{RS} = \hat{i} + \hat{j} - \hat{k}$ $\cos \theta = \frac{\overrightarrow{RQ} \cdot \overrightarrow{RS}}{ \overrightarrow{RQ} \overrightarrow{RS} } = \frac{ 1 - 8 - 2 }{\sqrt{69} \sqrt{3}} = \frac{9}{3\sqrt{23}}$ $\cos \theta = \frac{3}{\sqrt{23}}$ $RS = \sqrt{69} \times \frac{3}{23} = 3\sqrt{3}$ $\sin \theta = \frac{\sqrt{14}}{\sqrt{23}} = \frac{QS}{\sqrt{69}}$ $QS = \sqrt{42}$ $\text{Area} = \frac{1}{2} \cdot 2QS \cdot RS = \sqrt{42} \cdot 3\sqrt{3}$ $\lambda = 9\sqrt{14}$ $\lambda^2 = 81 \cdot 14 = 14K$ $K = 81$	
72.	Let 'm' be the real solution to the equation $29 \left(\frac{17-7x}{x+2} + \frac{8x+55}{x+3} \right) = 31 \left(\frac{24-5x}{x+1} + \frac{5-6x}{x+4} \right) + 370$ There are positive integers a, & b such that $m = -\frac{a}{b}$ then the minimum value of a + b is _____.	7

Sol.	$29\left(\frac{31}{x+2}-7+\frac{31}{x+3}+8\right)=31\left(\frac{29}{x+1}-5+\frac{29}{x+4}-6\right)+370$ $29 \times 31\left(\frac{1}{x+2}+\frac{1}{x+3}\right)=31 \times 29\left(\frac{1}{x+1}+\frac{1}{x+4}\right)$ $\frac{1}{x+2}+\frac{1}{x+3}=\frac{1}{x+1}+\frac{1}{x+4}$ $x^2+4x+3=x^2+6x+8$ $\Rightarrow x=-5/2 \quad \text{Ans: } 5+2=7.$	
73.	Let $I = \int \frac{(e^x-1)(\sin x - \cos x) + x \cos x}{\sin^2 x + (e^x - 1 - x)^2} dx = \tan^{-1}(f(x)) + C$ where 'C' is the constant of integration and $\lim_{x \rightarrow 0} f(x) = 0$, if $\lim_{x \rightarrow 0} \frac{f(x)}{x} = \frac{p}{q}$ where $p, q \in N$ then find the least value of $p + q$	3
Sol.	$I = \int \frac{(e^x-1)(\sin x - \cos x) + x \cos x}{\sin^2 x \left\{ 1 + \left(\frac{e^x - 1 - x}{\sin x} \right)^2 \right\}} dx$ Let $\frac{e^x - 1 - x}{\sin x} = t$, $I = \int \frac{1}{1+t^2} dt = \tan^{-1}\left(\frac{e^x - 1 - x}{\sin x}\right) + c$ $f(x) = \frac{e^x - 1 - x}{\sin x}$ $\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x \left(\frac{\sin x}{x} \right)} = \frac{1}{2} = \frac{p}{q}$ $\frac{(p+q)}{\text{least}} = 3.$	
74.	Let A_n be the sum of the first n terms of the geometric series $704 + \frac{704}{2} + \frac{704}{4} + \frac{704}{8} + \dots$ and B_n be the sum of the first terms of the geometric series. $1984 - \frac{1984}{2} + \frac{1984}{4} - \frac{1984}{8} + \dots$ if $A_n = B_n$. Then find the value of n (where $n \in N$).	5
Sol.	$A_n = 704 \frac{\left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} = 1408 \left(1 - \left(\frac{1}{2}\right)^n\right).$ $B_n = (1984) \left(\frac{2}{3}\right) \left(1 - \left(\frac{-1}{2}\right)^n\right)$ $A_n = B_n$ $33 \left(1 - \left(\frac{1}{2}\right)^n\right) = 31 \left(1 - \left(\frac{-1}{2}\right)^n\right)$	

75.	Four persons can hit a target correctly with probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ and $\frac{1}{8}$ respectively. If all hit at the target independently, then the probability that the target would be hit, is n . Then $192n$ is _____.	15 0
Sol.	$P \text{ (at least one hits the target)}$ $= 1 - P \text{ (none of them hits the target)}$ $= 1 - \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{8}\right)$ $= 1 - \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{7}{8} = 1 - \frac{7}{32} = \frac{25}{32}$	